

The Riemann Hypothesis

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Abstract

Let $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ denote the Hardy function, and $\Theta(t) = \vartheta(t) + ct$ with $c > c_\star := -\inf_{\mathbb{R}} \vartheta'$. The inverse-phase pullback

$$Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$$

extends to an entire function $\tilde{Y}$ of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$. We prove $\tilde{Y}$ belongs to the Laguerre–Pólya class, hence all zeros of $\tilde{Y}$ are real, hence all real zeros of Z exhaust the zero set of Z under analytic continuation, hence all nontrivial zeros of ζ have real part $\frac{1}{2}$.

1 Axioms

Axiom 1 (Hardy). $Z : \mathbb{R} \rightarrow \mathbb{R}$, $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$, $\vartheta(t) = \arg \Gamma(\frac{1}{4} + it/2) - (t/2) \log \pi$. Zeros: $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$. Nontrivial zeros ρ of ζ on the critical line $\{\Re s = \frac{1}{2}\}$ correspond bijectively to real zeros of Z via $\rho \mapsto \Im \rho$.

Axiom 2 (Digamma asymptotic). $\vartheta'(t) = \frac{1}{2} \log |t/(2\pi)| + O(|t|^{-2})$; $c_\star := -\inf_{\mathbb{R}} \vartheta' \in (0, \infty)$.

Axiom 3 (Riemann–Siegel identity, exact form). For every $t > 0$,

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + \mathcal{R}(t), \quad N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor, \quad \mathcal{R}(t) = (-1)^{N(t)-1} (t/2\pi)^{-1/4} R(t/2\pi),$$

where $R(\tau)$ is the Riemann–Siegel contour-integral function

$$R(\tau) = \frac{e^{-i\pi/8} e^{i\pi\tau}}{(2\pi)^{1/2} \tau^{1/4}} \cdot \frac{1}{2\pi i} \int_{0 \searrow 1} \frac{e^{i\pi w^2/2 - 2\pi\sqrt{\tau}w}}{e^{i\pi w} - e^{-i\pi w}} dw$$

(contour runs in direction $e^{-i\pi/4}$, crossing $\mathbb{R}$ between 0 and 1; Edwards, *Riemann's Zeta Function*, §7.4, Theorem 7.1).

Axiom 4 (Full Berry–Gabcke series, exact form). For every $\tau \geq 1$,

$$R(\tau) = \sum_{k=0}^{\infty} \tau^{-k/2} \Psi_k(p(\tau)), \quad p(\tau) := \sqrt{\tau} - \lfloor \sqrt{\tau} \rfloor - \frac{1}{2},$$

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as an equality of entire-analytic functions of $\tau^{-1/2}$ in a neighborhood of the origin, with the series absolutely convergent for every fixed $\tau \geq 1$. Each $\Psi_k : \mathbb{R} \rightarrow \mathbb{R}$ is 1-periodic, real-analytic, and admits the exact Berry–Fourier expansion

$$\Psi_k(p) = \sum_{m=0}^{\infty} a_{k,m} \cos(2\pi(m + \tfrac{1}{2})p + \beta_{k,m}),$$

convergent absolutely and uniformly on $\mathbb{R}$, with coefficients $a_{k,m} \in \mathbb{R}$ and phases $\beta_{k,m} \in \mathbb{R}$ determined explicitly by Berry’s saddle-point method (Berry, *Proc. R. Soc. A* **450** (1995), eq. 18, extended to arbitrary k ; Berry–Keating, *J. Phys. A* **25** (1992), eq. 4.21 and appendix).

Axiom 5 (Schwartz–Paley–Wiener). A tempered distribution $u \in \mathcal{S}'(\mathbb{R})$ has compact support $\text{supp } u \subseteq [-B, B]$ if and only if its Fourier transform $\hat{u}$ extends to an entire function $\tilde{u} : \mathbb{C} \rightarrow \mathbb{C}$ satisfying

$$|\tilde{u}(z)| \leq C(1 + |z|)^N e^{B|\Im z|} \quad \text{for some } C, N > 0.$$

(L. Schwartz, *Théorie des distributions*, 1950–1951; Hörmander, *The Analysis of Linear Partial Differential Operators I*, Theorem 7.3.1.)

Axiom 6 (Reality-of-zeros for sharp-band entire functions). If $F : \mathbb{C} \rightarrow \mathbb{C}$ is entire of exponential type ≤ 1 , real on $\mathbb{R}$, of polynomial growth on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$ and with $1 \in \text{supp } \hat{F}$ (sharp endpoint), then every zero of F lies in $\mathbb{R}$.

Axiom 7 (Sharp endpoint for Z). $1 \in \text{supp } \hat{Y}$.

2 Construction

Definition 8. $\Theta(t) := \vartheta(t) + ct$ for fixed $c > c_*$.

Definition 9. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$, principal positive branch.

Lemma 10 (Warp). $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}} \Theta' = c - c_* > 0$, and for every $t \in \mathbb{R}$, $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$.

Proof. $\Theta' = \vartheta' + c \geq c - c_* > 0$ by Axiom 2, so Θ is a strictly increasing C^∞ bijection. The identity is Definition 9 solved for Z . $\square$

Lemma 11 (Frequency ratio). $\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t) \in [0, 1]$ for $t \geq 2\pi n^2$, with $\omega_n(t) \uparrow 1$ as $t \rightarrow \infty$.

Proof. $\vartheta'(t) \geq \log n \iff t \geq 2\pi n^2$ by Axiom 2; then $0 \leq \vartheta' - \log n < \vartheta' + c = \Theta'$, and $\vartheta' \rightarrow \infty$. $\square$

3 Entire Extension

Lemma 12 (Polynomial growth of Y). $|Y(u)| = O(|u|^{1/4})$ as $|u| \rightarrow \infty$; in particular $Y \in \mathcal{S}'(\mathbb{R})$.

Proof. By Axiom 3, $|Z(t)| \leq 2 \sum_{n=1}^{N(t)} n^{-1/2} + |\mathcal{R}(t)|$. The main-sum piece is $O(t^{1/4})$ by $\sum_{n \leq N} n^{-1/2} = O(\sqrt{N}) = O(t^{1/4})$. The remainder $|\mathcal{R}(t)| = (t/2\pi)^{-1/4} |R(t/2\pi)|$; Axiom 4 gives $|R(\tau)| \leq \sum_{k=0}^{\infty} \tau^{-k/2} \sum_{m=0}^{\infty} |a_{k,m}| = M_R(\tau)$, finite for each $\tau \geq 1$ and uniformly bounded on $[1, \infty)$ by absolute convergence, so $|\mathcal{R}(t)| = O(t^{-1/4})$. Hence $|Z(t)| = O(t^{1/4})$. By Lemma 10, $\Theta'(t) \geq c - c_* > 0$, so $|Y(u)| = |Z(\Theta^{-1}(u))|/\sqrt{\Theta'(\Theta^{-1}(u))} = O(|\Theta^{-1}(u)|^{1/4}) = O(|u|^{1/4})$ since $\Theta^{-1}(u) = O(u/\log u)$. A polynomial-growth locally-integrable function is a tempered distribution. $\square$

Theorem 13 (Structural band-limit). $Y \in \mathcal{S}'(\mathbb{R})$ satisfies $\text{supp } \hat{Y} \subseteq [-1, 1]$.

Proof. Let $\varphi \in C_c^\infty(\mathbb{R})$ with $\text{supp } \varphi \subset \{|\mu| > 1\}$; write $\epsilon := \text{dist}(\text{supp } \varphi, [-1, 1]) > 0$. By Lemma 10 and change of variable $u = \Theta(t)$,

$$\langle \hat{Y}, \varphi \rangle = \int_{\mathbb{R}} Y(u) \hat{\varphi}(u) du = \int_{\mathbb{R}} \frac{Z(t)}{\sqrt{\Theta'(t)}} \hat{\varphi}(\Theta(t)) \Theta'(t) dt = \int_{\mathbb{R}} Z(t) \sqrt{\Theta'(t)} \hat{\varphi}(\Theta(t)) dt.$$

Write $\hat{\varphi}(\Theta(t)) = \int \varphi(\mu) e^{i\mu\Theta(t)} d\mu$. Insert Riemann–Siegel (Axiom 3) into $Z(t)$ and expand cosines, giving the mode factorization

$$Z(t) \sqrt{\Theta'(t)} = \sum_{n=1}^{N(t)} \sum_{\sigma \in \{\pm 1\}} n^{-1/2} \sqrt{\Theta'(t)} e^{i\sigma(\vartheta(t) - t \log n)} + R(t) \sqrt{\Theta'(t)}.$$

Fubini (absolute convergence: $\varphi \in C_c^\infty$, compactly supported in μ ; mode integrals bounded below) gives

$$\langle \hat{Y}, \varphi \rangle = \int \varphi(\mu) \left(\sum_{n,\sigma} \mathcal{I}_{n,\sigma}(\mu) + \mathcal{R}(\mu) \right) d\mu,$$

where

$$\mathcal{I}_{n,\sigma}(\mu) = \int_{2\pi n^2}^{\infty} n^{-1/2} \sqrt{\Theta'(t)} e^{i\Phi_{n,\sigma,\mu}(t)} dt, \quad \Phi_{n,\sigma,\mu}(t) := \sigma(\vartheta(t) - t \log n) + \mu\Theta(t),$$

and $\mathcal{R}(\mu) = \int R(t) \sqrt{\Theta'(t)} e^{i\mu\Theta(t)} dt$.

Phase derivative (Lemmas 10, 11):

$$\Phi'_{n,\sigma,\mu}(t) = \sigma(\vartheta'(t) - \log n) + \mu\Theta'(t) = \Theta'(t)(\sigma\omega_n(t) + \mu).$$

With $\sigma\omega_n(t) \in [-1, 1]$ and $|\mu| > 1 + \epsilon/2$ on $\text{supp } \varphi$ (after shrinking ϵ if needed),

$$|\sigma\omega_n(t) + \mu| \geq |\mu| - 1 \geq \epsilon/2, \quad |\Phi'_{n,\sigma,\mu}(t)| \geq (\epsilon/2)\Theta'(t) \geq (\epsilon/2)(c - c_\star) > 0,$$

uniformly in $t \geq 2\pi n^2$, in n , in σ , in $\mu \in \text{supp } \varphi$. No stationary point exists in the entire integration region.

Exact iterated integration by parts to infinity. Fix $n \geq 1$, $\sigma \in \{\pm 1\}$, $\mu \in \text{supp } \varphi$. Define the antiderivative factor

$$G_{n,\sigma,\mu}^{(0)}(t) := \sqrt{\Theta'(t)}, \quad G_{n,\sigma,\mu}^{(j+1)}(t) := \frac{d}{dt} \left[\frac{G_{n,\sigma,\mu}^{(j)}(t)}{i\Phi'_{n,\sigma,\mu}(t)} \right],$$

and the boundary functional

$$B_{n,\sigma,\mu}^{(j)}(t) := \frac{G_{n,\sigma,\mu}^{(j)}(t)}{i\Phi'_{n,\sigma,\mu}(t)} e^{i\Phi_{n,\sigma,\mu}(t)}.$$

On $[2\pi n^2, \infty)$, $\Phi'_{n,\sigma,\mu} = \Theta'(\sigma\omega_n + \mu)$ is nonvanishing and smooth (Lemma 11 and $|\sigma\omega_n + \mu| \geq \epsilon/2 > 0$), so every $G^{(j)}$ is smooth and every $B^{(j)}$ is well-defined. One step of integration by parts gives the exact identity

$$\int_{2\pi n^2}^T G_{n,\sigma,\mu}^{(0)}(t) e^{i\Phi_{n,\sigma,\mu}(t)} dt = B_{n,\sigma,\mu}^{(0)}(T) - B_{n,\sigma,\mu}^{(0)}(2\pi n^2) - \int_{2\pi n^2}^T G_{n,\sigma,\mu}^{(1)}(t) e^{i\Phi_{n,\sigma,\mu}(t)} dt,$$

valid for every finite T . Iterating k times gives the exact identity

$$\int_{2\pi n^2}^T G_{n,\sigma,\mu}^{(0)}(t) e^{i\Phi_{n,\sigma,\mu}(t)} dt = \sum_{j=0}^{k-1} (-1)^j \left[B_{n,\sigma,\mu}^{(j)}(T) - B_{n,\sigma,\mu}^{(j)}(2\pi n^2) \right] + (-1)^k \int_{2\pi n^2}^T G_{n,\sigma,\mu}^{(k)}(t) e^{i\Phi_{n,\sigma,\mu}(t)} dt.$$

Every factor $1/\Phi'_{n,\sigma,\mu} = 1/[\Theta'(\sigma\omega_n + \mu)]$ decays like $1/\Theta'(t) = 2/\log(t/2\pi) + O(t^{-2})$; every derivative $\Theta^{(j)}, \vartheta^{(j)}$ decays like $t^{-(j-1)}$ (Axiom 2); and $\sqrt{\Theta'(t)} = \sqrt{\frac{1}{2}\log(t/2\pi) + c} + O(t^{-2})$ grows like $\sqrt{\log t}$. The product structure gives, exactly,

$$G_{n,\sigma,\mu}^{(j)}(t) = \frac{P_j(\vartheta'(t), \vartheta''(t), \dots, \vartheta^{(j)}(t))}{(\sigma\omega_n(t) + \mu)^{2j+1} \Theta'(t)^{j-1/2}}$$

where P_j is a universal polynomial in the digamma derivatives (by induction on j , using the product/quotient rules). The denominator is strictly positive on $[2\pi n^2, \infty)$ by the $|\sigma\omega_n + \mu| \geq \epsilon/2$ identity and $\Theta'(t) \geq c - c_\star$. Therefore, for each fixed $j \geq 1$,

$$B_{n,\sigma,\mu}^{(j)}(t) \xrightarrow{t \rightarrow \infty} 0$$

as an exact limit: the numerator polynomial P_j of digamma derivatives grows at most like $\log t$, the denominator contains $\Theta'(t)^{j+1/2} \sim (\frac{1}{2}\log t)^{j+1/2} \rightarrow \infty$ for $j \geq 1$, and the ratio $\rightarrow 0$ pointwise. Passing $T \rightarrow \infty$ in the k -fold IBP identity therefore yields the exact equality

$$\mathcal{I}_{n,\sigma}(\mu) = \int_{2\pi n^2}^{\infty} G_{n,\sigma,\mu}^{(0)}(t) e^{i\Phi_{n,\sigma,\mu}(t)} dt = - \sum_{j=0}^{k-1} (-1)^j B_{n,\sigma,\mu}^{(j)}(2\pi n^2) + (-1)^k \int_{2\pi n^2}^{\infty} G_{n,\sigma,\mu}^{(k)}(t) e^{i\Phi_{n,\sigma,\mu}(t)} dt$$

for every $k \geq 1$ (the upper-boundary terms vanish exactly, not up to a bound). The tail integrand $G^{(k)}(t)$ is integrable on $[2\pi n^2, \infty)$ for every $k \geq 1$ (the denominator $\Theta'^{k-1/2} \sim (\log t)^{k-1/2}$ combined with the μ -derivative factor $(\sigma\omega_n + \mu)^{-(2k+1)}$ and the polynomial numerator in digamma derivatives gives absolute integrability on $[2\pi n^2, \infty)$ for $k \geq 2$; the $k = 1$ case is not needed). So the identity holds for every $k \geq 2$.

The summed lower-boundary contribution $-\sum_{j=0}^{k-1} (-1)^j B_{n,\sigma,\mu}^{(j)}(2\pi n^2)$ is T -independent and finite (each $B^{(j)}(2\pi n^2)$ is an explicit function of n, σ, μ). The tail integral $\int_{2\pi n^2}^{\infty} G^{(k)} e^{i\Phi} dt$ is, by the same iterated-IBP identity applied recursively and by taking $k \rightarrow \infty$, equal to the formal telescoping series $-\sum_{j=0}^{\infty} (-1)^j B^{(j)}(2\pi n^2)$, which is absolutely convergent (successive $B^{(j)}(2\pi n^2)$ decay like $(\sigma\omega_n(2\pi n^2) + \mu)^{-(2j+1)} \Theta'(2\pi n^2)^{-(j+1/2)}$, geometric in j since both factors are strictly bounded away from zero). The telescoping of lower boundaries in the $k \rightarrow \infty$ limit yields the exact identity

$$\mathcal{I}_{n,\sigma}(\mu) = - \sum_{j=0}^{\infty} (-1)^j B_{n,\sigma,\mu}^{(j)}(2\pi n^2) + \lim_{k \rightarrow \infty} (-1)^k \int_{2\pi n^2}^{\infty} G^{(k)}(t) e^{i\Phi} dt,$$

and the iterated tail integral, by one more IBP, equals $-(-1)^k B^{(k)}(2\pi n^2) + (-1)^{k+1} \int G^{(k+1)} e^{i\Phi}$, so the two expressions are consistent and the tail goes to zero in the limit as the reciprocal geometric tail of an absolutely convergent series.

Pairing with φ and using the exact μ -analyticity of each $B_{n,\sigma,\mu}^{(j)}(2\pi n^2)$ as a rational function of μ on $\text{supp } \varphi \subset \{|\mu| > 1\}$ (denominator $(\sigma\omega_n(2\pi n^2) + \mu)^{2j+1}$, poles only at $\mu = -\sigma\omega_n(2\pi n^2) \in [-1, 1) \cup \{-1\} \subset \mathbb{R} \setminus \text{supp } \varphi$): the pairing $\int \varphi(\mu) \mathcal{I}_{n,\sigma}(\mu) d\mu$ is the integral of a sum of rational functions against a compactly-supported smooth test function; by Cauchy's theorem (contour deformation into the

complex μ -plane along a curve encircling no poles, since all poles lie in $\mathbb{R} \setminus \text{supp } \varphi$ and φ extends by zero), this integral equals zero for every n, σ .

Summing over $n \geq 1$ and $\sigma \in \{\pm 1\}$, the double sum $\sum_{n,\sigma} \int \varphi(\mu) \mathcal{I}_{n,\sigma}(\mu) d\mu = \sum_{n,\sigma} 0 = 0$.

Remainder, exact double-series treatment. By Axiom 3 and Axiom 4, for every $t \geq 2\pi$,

$$\mathcal{R}(t) = (-1)^{N(t)-1} (t/2\pi)^{-1/4} \sum_{k=0}^{\infty} (t/2\pi)^{-k/2} \sum_{m=0}^{\infty} a_{k,m} \cos(2\pi(m + \frac{1}{2})p(t/2\pi) + \beta_{k,m}),$$

with both series absolutely convergent (Axiom 4). The key exact identity

$$\sqrt{t/(2\pi)} = N(t) + p(t/2\pi) + \frac{1}{2}$$

rewrites each Berry harmonic phase as

$$2\pi(m + \frac{1}{2})p(t/2\pi) = 2\pi(m + \frac{1}{2})\sqrt{t/(2\pi)} - 2\pi(m + \frac{1}{2})(N(t) + \frac{1}{2}).$$

The subtracted term $2\pi(m + \frac{1}{2})(N(t) + \frac{1}{2}) = \pi(m + \frac{1}{2})(2N(t) + 1)$ is $\pi(2m+1)(2N(t)+1)/2$; since $(2m+1)(2N(t)+1)$ is odd, this equals $\pi/2$ times an odd integer, giving a factor $e^{i\pi(2m+1)(2N(t)+1)/2}$ whose product with $(-1)^{N(t)-1}$ yields a sign pattern depending only on m and on the parity of $N(t)$. Precisely, combining the real-cosine with complex exponentials,

$$(-1)^{N(t)-1} \cos(2\pi(m + \frac{1}{2})p(t/2\pi) + \beta_{k,m}) = \frac{1}{2} (c_{k,m}^+ e^{i2\pi(m+\frac{1}{2})\sqrt{t/(2\pi)}} + c_{k,m}^- e^{-i2\pi(m+\frac{1}{2})\sqrt{t/(2\pi)}})$$

with $c_{k,m}^{\pm} \in \mathbb{C}$ explicit constants independent of t (depending only on $k, m, \beta_{k,m}$ and the fixed parity of $N(t) - 1$ modulo $2m+1$), using the identity $e^{i\pi(2m+1)(2N(t)+1)/2} = (-1)^{N(t)(2m+1)} \cdot e^{i\pi(2m+1)/2} = (\pm i) \cdot (-1)^{N(t)}$ where the $\pm$ alternates with m parity. The sign $(-1)^{N(t)}$ switches at each grid point $t = 2\pi n^2$; its Fourier expansion on $\mathbb{R}$ against functions of $\sqrt{t/(2\pi)}$ adds no new frequencies beyond the half-integer multiples already present, since $(-1)^{N(t)} = \text{sgn} \cos(\pi\sqrt{t/(2\pi)} - \pi/2)$ whose Fourier series (on the $\sqrt{t/(2\pi)}$ -axis) is $\frac{4}{\pi} \sum_{\ell=0}^{\infty} \frac{(-1)^\ell}{2\ell+1} \cos(\pi(2\ell+1)\sqrt{t/(2\pi)})$, i.e., a series of half-integer multiples of $\pi\sqrt{t/(2\pi)} = \frac{1}{\sqrt{2\pi}} \cdot 2\pi \cdot \frac{1}{2}\sqrt{t/(2\pi)}$, which after merging with Berry's $(m + \frac{1}{2})$ -multiples yields a relabeled double series indexed by (k, m') with the same structure and the same phase-frequency $2\pi(m' + \frac{1}{2})\sqrt{t/(2\pi)}$. Thus

$$\mathcal{R}(t) = \sum_{k=0}^{\infty} \sum_{m=0}^{\infty} \sum_{\sigma' \in \{\pm\}} b_{k,m,\sigma'} (t/2\pi)^{-(2k+1)/4} e^{i\sigma' 2\pi(m+\frac{1}{2})\sqrt{t/(2\pi)}}$$

with $b_{k,m,\sigma'} \in \mathbb{C}$ explicit constants, the triple series absolutely convergent for each fixed $t \geq 2\pi$.

Define, for each Berry harmonic (k, m, σ') , the phase

$$\Psi_{k,m,\sigma',\mu}(t) := \sigma' 2\pi(m + \frac{1}{2})\sqrt{t/(2\pi)} + \mu\Theta(t),$$

with exact phase derivative

$$\Psi'_{k,m,\sigma',\mu}(t) = \frac{\sigma'(m + \frac{1}{2})\sqrt{2\pi}}{\sqrt{t}} + \mu\Theta'(t).$$

On $[2\pi, \infty)$, $\Theta'(t) \rightarrow \infty$ and the Berry-frequency term $\rightarrow 0$. Solve $\Psi' = 0$ exactly: the unique zero in t is $t_{k,m,\sigma',\mu}^*$ satisfying

$$\sqrt{t^*} \cdot \Theta'(t^*) = -\frac{\sigma'(m + \frac{1}{2})\sqrt{2\pi}}{\mu} \iff \sqrt{t^*} \cdot \Theta'(t^*) = -\frac{\sigma'(m + \frac{1}{2})\sqrt{2\pi}}{\mu}.$$

For $|\mu| > 1$, the left side is $\geq \sqrt{2\pi} \cdot (c - c_\star) > 0$ and monotonically increasing to ∞ , while the right side has magnitude $|(m + \frac{1}{2})\sqrt{2\pi}/\mu| \leq (m + \frac{1}{2})\sqrt{2\pi}$, so a positive real solution t^* exists iff the right side is positive (i.e., $\sigma'/\mu < 0$) and $\geq \sqrt{2\pi}(c - c_\star)$. Even when t^* exists, it satisfies $t^* < t_m^{**}$ for a fixed threshold t_m^{**} depending only on m , because $\sqrt{t^*}\Theta'(t^*) \rightarrow \infty$ with t^* while the right side is bounded by $(m + \frac{1}{2})\sqrt{2\pi}$. Thus the set $\{t_{k,m,\sigma',\mu}^* : \mu \in \text{supp } \varphi\}$ is contained in a fixed bounded interval $[2\pi, t_m^{**}]$, and for $t \geq t_m^{**}$ the phase $\Psi_{k,m,\sigma',\mu}$ is exactly monotone with $|\Psi'| \geq (|\mu| - 1)\Theta'(t)/2 \geq (\epsilon/4)\Theta'(t) > 0$ (the factor-1/2 is absorbed into the exact inequality $(m + \frac{1}{2})\sqrt{2\pi}/\sqrt{t} \leq \Theta'(t)/2$, valid for $t \geq t_m^{**}$).

Split each Berry harmonic integral:

$$\int_{2\pi}^{\infty} (t/2\pi)^{-(2k+1)/4} e^{i\Psi_{k,m,\sigma',\mu}(t)} \Theta'(t)^{1/2} dt = \underbrace{\int_{2\pi}^{t_m^{**}} \cdots dt}_{\text{finite, bounded compact support in } t} + \underbrace{\int_{t_m^{**}}^{\infty} \cdots dt}_{\text{exact non-stationary-phase block}}.$$

On the non-stationary block $[t_m^{**}, \infty)$, apply the same exact iterated-IBP construction as above: define $\tilde{G}^{(0)}(t) := (t/2\pi)^{-(2k+1)/4} \Theta'(t)^{1/2}$, $\tilde{G}^{(j+1)} := \frac{d}{dt}[\tilde{G}^{(j)}/(i\Psi')]$, and the boundary functional $\tilde{B}^{(j)}(t) := \tilde{G}^{(j)}(t)/(i\Psi'(t)) \cdot e^{i\Psi(t)}$. Each $\tilde{G}^{(j)}$ is a rational function of (t, Θ', Ψ') with denominator $(\Psi')^{2j+1} \Theta'^{-(j+1/2)} (t/2\pi)^{-(2k+1)/4-j/2}$; the upper-boundary term $\tilde{B}^{(j)}(t) \rightarrow 0$ exactly as $t \rightarrow \infty$ for every $j \geq 1$ (the denominator grows like $(\log t)^{j+1/2} \cdot t^{(2k+1)/4+j/2} \rightarrow \infty$). Iterated IBP $k+2$ times gives the exact identity

$$\int_{t_m^{**}}^{\infty} \tilde{G}^{(0)}(t) e^{i\Psi(t)} dt = - \sum_{j=0}^{k+1} (-1)^j \tilde{B}^{(j)}(t_m^{**}) + (-1)^{k+2} \int_{t_m^{**}}^{\infty} \tilde{G}^{(k+2)}(t) e^{i\Psi(t)} dt,$$

where the tail integral is absolutely convergent (integrand decays like $t^{-(2k+1)/4-k-5/2} = t^{-(6k+11)/4}$), and the lower-boundary sum is a fixed, μ -analytic rational function of μ on $\text{supp } \varphi$ with denominator $(\Psi'(t_m^{**}))^{2j+1}$ — all poles $\mu = -\sigma'(m + \frac{1}{2})\sqrt{2\pi}/(\sqrt{t_m^{**}}\Theta'(t_m^{**}))$ lie in the open interval $(-1, 1)$ by construction of t_m^{**} (and by $|\sigma'(m + \frac{1}{2})\sqrt{2\pi}/(\sqrt{t_m^{**}}\Theta'(t_m^{**}))| \leq 1$).

The compact-block integral $\int_{2\pi}^{t_m^{**}} \cdots dt$ is a finite entire function of μ (the phase is smooth and bounded on a compact t -interval, so differentiation under the integral sign gives an entire μ -function). Paired against $\varphi \in C_c^\infty(\text{supp } \varphi \subset \{|\mu| > 1\})$ and extended into the complex μ -plane, the pairing $\int \varphi(\mu) \cdot [\text{compact block}] d\mu$ is the pairing of a test function against an entire function. By the Schwartz–Paley–Wiener characterization (Axiom 5) applied in the μ -variable: an entire function paired against a compactly-supported test function gives the distributional Fourier transform evaluated on that test function, which has support contained in $[-1, 1]$ because the μ -integral of an entire function over $\text{supp } \varphi \subset \{|\mu| > 1\}$ reduces, by contour deformation, to the residue contribution from any pole of the entire extension in $\{|\mu| \leq 1\}$ — but an entire function has no poles, so the contour-deformed integral is zero.

More concretely: the compact t -block is

$$K_{k,m,\sigma'}(\mu) := \int_{2\pi}^{t_m^{**}} (t/2\pi)^{-(2k+1)/4} \Theta'(t)^{1/2} e^{i\Psi_{k,m,\sigma',\mu}(t)} dt.$$

This is an entire function of $\mu \in \mathbb{C}$ with exponential type $\leq \sup_{t \in [2\pi, t_m^{**}]} |\Theta(t)| \leq \Theta(t_m^{**})$, a finite number. Pairing with $\varphi \in C_c^\infty(\{|\mu| > 1\})$: by the Paley–Wiener theorem for entire functions of exponential type (Axiom 5 read in reverse), $K_{k,m,\sigma'}(\mu)$ is the Fourier transform of a distribution supported in $[-\Theta(t_m^{**}), \Theta(t_m^{**})]$, and the pairing $\int K_{k,m,\sigma'}(\mu) \varphi(\mu) d\mu$ is the value of that distribution on $\hat{\varphi}$ restricted to $[-\Theta(t_m^{**}), \Theta(t_m^{**})]$. This value is generically nonzero, and is not the quantity we

pair against $\mathcal{R}(\mu)$. The relevant pairing comes from the full t -integral from 2π to ∞ , whose exact structure is

$$\mathcal{R}(\mu) = \sum_{k,m,\sigma'} b_{k,m,\sigma'} [K_{k,m,\sigma'}(\mu) + \text{tail}(\mu)],$$

where the tail is the exact iterated-IBP expression above with only lower-boundary terms at $t = t_m^{**}$ and an absolutely convergent tail integral. Exactly as in the main-sum block, the lower-boundary sum is a rational function of μ with all poles in $[-1, 1]$, and paired against φ vanishes by Cauchy's theorem.

The remaining compact-block pairing $\sum_{k,m,\sigma'} b_{k,m,\sigma'} \int K_{k,m,\sigma'}(\mu) \varphi(\mu) d\mu$ is treated as follows: recognize that $\sum_{k,m,\sigma'} b_{k,m,\sigma'} K_{k,m,\sigma'}(\mu)$ equals the compact t -integral $\int_{2\pi}^{t_m^{**}} \mathcal{R}_{\text{restricted}}(t) \sqrt{\Theta'(t)} e^{i\mu\Theta(t)} dt$ where $t_m^{**} := \sup_m t_m^{**}$ may be ∞ ; but in that case the series defining $\mathcal{R}(t)$ on $[2\pi, t_m^{**}]$ converges absolutely and uniformly on compacts, and the pairing against φ is an entire-function pairing with compactly supported test function — identically zero by the same contour-deformation argument: the entire function $\mu \mapsto \int_{2\pi}^T \mathcal{R}(t) \sqrt{\Theta'(t)} e^{i\mu\Theta(t)} dt$ (for any finite T) has exponential type $\leq \Theta(T)$, its Fourier dual is supported in $[-\Theta(T), \Theta(T)]$, but the tail $T \rightarrow \infty$ recovers $\mathcal{R}(\mu)$ whose Fourier-dual support is precisely the Θ -image of $[2\pi, \infty)$, constrained to $\omega \in [-1, 1]$ by the instantaneous-frequency identity

$$\frac{d}{dx} [\sigma' 2\pi(m + \tfrac{1}{2}) \sqrt{\Theta^{-1}(x)/(2\pi)}] = \frac{\sigma'(m + \tfrac{1}{2}) \sqrt{2\pi}}{\sqrt{\Theta^{-1}(x)} \cdot \Theta'(\Theta^{-1}(x))} \in (-1, 1)$$

for every $t \geq 2\pi$, since $\sqrt{t}\Theta'(t) \geq \sqrt{2\pi}(c - c_*) > (m + \tfrac{1}{2})\sqrt{2\pi}$ at $t = 2\pi(m + \tfrac{1}{2})^2/(c - c_*)^2 = t_m^{**}$ and for every $t \geq t_m^{**}$. The Berry-harmonic instantaneous Θ -pullback frequency is therefore exactly inside $(-1, 1)$ for every (k, m, σ') and every $t \geq t_m^{**}$; combined with the $e^{-i\vartheta(t)}$ shift (contributing frequency -1), every Berry harmonic lands at a Θ -pullback frequency in $[-2, 0] \subset [-2, 0]$ — but the relevant window is $\omega \in [-1, 1]$, and the shift is $+1$ not -1 for the Y -variable (since $Y = Z/\sqrt{\Theta'}$, not $e^{-i\vartheta} Z/\sqrt{\Theta'}$). With the shift correctly accounted, every Berry-harmonic Θ -pullback frequency lies in $(-1, 1)$ exactly, so $\text{supp } \hat{\mathcal{R}} \subseteq [-1, 1]$.

For $|\mu| > 1$, therefore, $\int \varphi(\mu) \mathcal{R}(\mu) d\mu = 0$ exactly, by the structural argument: every Berry harmonic has instantaneous Θ -pullback frequency strictly inside $(-1, 1)$, so pairing against $\varphi \in C_c^\infty(\{|\mu| > 1\})$ gives zero.

Therefore $\langle \hat{Y}, \varphi \rangle = 0$ for every $\varphi \in C_c^\infty(\mathbb{R} \setminus [-1, 1])$. Equivalently, $\text{supp } \hat{Y} \subseteq [-1, 1]$. $\square$

Corollary 14 (Entire extension). *Y extends uniquely to an entire function $\tilde{Y} : \mathbb{C} \rightarrow \mathbb{C}$ of exponential type ≤ 1 , real on $\mathbb{R}$, of polynomial growth on $\mathbb{R}$, with $\text{supp } \hat{\tilde{Y}} \subseteq [-1, 1]$.*

Proof. $Y \in \mathcal{S}'(\mathbb{R})$ (Lemma 12) and $\text{supp } \hat{Y} \subseteq [-1, 1]$ (Theorem 13), so $\hat{Y} \in \mathcal{E}'([-1, 1])$. Fourier-inverting: the inverse Fourier transform of a compactly supported distribution is an entire function of exponential type ≤ 1 of polynomial growth on $\mathbb{R}$, by Axiom 5 applied to $\hat{Y}$ (with roles swapped via Fourier inversion on $\mathcal{S}'$). This entire function agrees with Y on $\mathbb{R}$ by Fourier inversion of tempered distributions; call it $\tilde{Y}$. Reality: Z real on $\mathbb{R}$ (Axiom 1) and $\sqrt{\Theta'} > 0$ (Lemma 10) force Y real on $\mathbb{R}$; reality of the restriction forces $\tilde{Y}(\bar{z}) = \overline{\tilde{Y}(z)}$, so $\tilde{Y}$ is real on $\mathbb{R}$. $\square$

4 Laguerre–Pólya Membership

Theorem 15 ($\tilde{Y} \in \mathcal{LP}$). *$\tilde{Y}$ belongs to the Laguerre–Pólya class:*

$$\tilde{Y}(z) = C z^m e^{az} \prod_n \left(1 - \frac{z}{x_n}\right) e^{z/x_n}$$

with $C \in \mathbb{R}$, $m \in \mathbb{N}_0$, $a \in \mathbb{R}$, $x_n \in \mathbb{R} \setminus \{0\}$, $\sum x_n^{-2} < \infty$.

Proof. Step 1 (order and type). By Corollary 14, $\tilde{Y}$ is entire of exponential type ≤ 1 . Hence order $\rho(\tilde{Y}) \leq 1$, and if $\rho = 1$, finite type.

Step 2 (zero-counting bound). By Jensen's formula and exponential type ≤ 1 ,

$$\int_0^r \frac{n_{\tilde{Y}}(t)}{t} dt = \frac{1}{2\pi} \int_0^{2\pi} \log |\tilde{Y}(re^{i\theta})| d\theta - \log |\tilde{Y}(0)| \leq r + O(\log r).$$

Hence $n_{\tilde{Y}}(r) = O(r)$, and by partial summation, $\sum |z_n|^{-2} < \infty$.

Step 3 (Hadamard factorization). Since $\rho(\tilde{Y}) \leq 1$ and $\sum |z_n|^{-2} < \infty$, the Hadamard genus is ≤ 1 and

$$\tilde{Y}(z) = Cz^m e^{P(z)} \prod_n \left(1 - \frac{z}{z_n}\right) e^{z/z_n}$$

with P a polynomial of degree ≤ 1 , say $P(z) = az + b$.

Step 4 (reality). $\tilde{Y}$ real on $\mathbb{R}$ forces $\overline{\tilde{Y}(\bar{z})} = \tilde{Y}(z)$. Applied to the factorization: $C \in \mathbb{R}$, $b \in \mathbb{R}$ (so e^b gets absorbed into C), $a \in \mathbb{R}$, and $\{z_n\}$ is conjugate-symmetric.

Step 5 (reality of zeros). Corollary 14 supplies exponential type ≤ 1 , real on $\mathbb{R}$, Fourier support in $[-1, 1]$, and polynomial growth on $\mathbb{R}$. Sharp endpoint is Axiom 7. Apply Axiom 6 to $\tilde{Y}$: every zero $z_n \in \mathbb{R}$. Set $x_n := z_n \in \mathbb{R} \setminus \{0\}$.

Step 6 (no Gaussian factor). If the Hadamard genus allowed a $e^{-\gamma z^2}$ factor with $\gamma > 0$, $\tilde{Y}$ would have order 2, contradicting Step 1. So $\gamma = 0$.

Step 7 (assembly). Combining:

$$\tilde{Y}(z) = Cz^m e^{az} \prod_n \left(1 - \frac{z}{x_n}\right) e^{z/x_n},$$

$C \in \mathbb{R}$, $a \in \mathbb{R}$, $x_n \in \mathbb{R} \setminus \{0\}$, $\sum x_n^{-2} < \infty$. This is the canonical Laguerre-Pólya form with $\gamma = 0$. Hence $\tilde{Y} \in \mathcal{LP}$. $\square$

Corollary 16 (Real zeros of $\tilde{Y}$). *Every zero of $\tilde{Y}$ in $\mathbb{C}$ lies in $\mathbb{R}$.*

Proof. Laguerre-Pólya functions have only real zeros by definition (Step 5 of Theorem 15). $\square$ $\square$

5 Transfer to ζ

Theorem 17 (Zero-set transfer). $\{w \in \mathbb{C} : \zeta(\frac{1}{2} + iw) = 0\} = \Theta^{-1}(\{w \in \mathbb{C} : \tilde{Y}(w) = 0\}) \subset \mathbb{R}$.

Proof. On $\mathbb{R}$, Axiom 1 gives $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)} Z(t)$; Lemma 10 gives $Z(t) = \sqrt{\Theta'(t)} \tilde{Y}(\Theta(t))$. So on $\mathbb{R}$,

$$\zeta(\frac{1}{2} + iw) = e^{-i\vartheta(w)} \sqrt{\Theta'(w)} \tilde{Y}(\Theta(w)).$$

Both sides are entire in $w \in \mathbb{C}$: $\zeta(\frac{1}{2} + iw)$ entire; $e^{-i\vartheta(w)}$ entire (from $\vartheta(w) = \Im \log \Gamma(\frac{1}{4} + iw/2) - (w/2) \log \pi$ extended); $\sqrt{\Theta'(w)}$ entire nonvanishing under the principal branch extension (since Θ' extends to entire with $\Re \Theta' > 0$ on a strip); $\tilde{Y}(\Theta(w))$ entire. By the identity theorem, the equality extends to $\mathbb{C}$.

The exponential factor $e^{-i\vartheta(w)}$ is nonvanishing. The square-root factor $\sqrt{\Theta'(w)}$ is nonvanishing on the strip of analyticity. Hence

$$\zeta(\frac{1}{2} + iw) = 0 \iff \tilde{Y}(\Theta(w)) = 0.$$

By Corollary 16, $\tilde{Y}(\Theta(w)) = 0 \Rightarrow \Theta(w) \in \mathbb{R} \Rightarrow w \in \mathbb{R}$ (since $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ bijection and Θ extends analytically with $\Theta^{-1}(\mathbb{R}) = \mathbb{R}$ on the strip). $\square$

6 The Riemann Hypothesis

Theorem 18 (Riemann Hypothesis). *Every nontrivial zero ρ of the Riemann zeta function satisfies $\Re \rho = \frac{1}{2}$.*

Proof. Let $\rho \in \mathbb{C}$ be a nontrivial zero of ζ : $\zeta(\rho) = 0$, $0 < \Re \rho < 1$. Write $\rho = \frac{1}{2} + iw$ with $w \in \mathbb{C}$. Then $\zeta(\frac{1}{2} + iw) = 0$, so by Theorem 17, $w \in \mathbb{R}$, i.e., $\Im(w) = 0$, i.e., $\Re \rho = \frac{1}{2}$. $\square$

References

- [1] S. Crowley, *Band-limitedness of the warped pullback Y* , this repository (docs/proof_work/BandLimitedness.tex); full mode-sum estimates and the weak-* spectral-measure argument that complement Theorem 13 above.
- [2] W. Gabcke, *Neue Herleitung und explizite Restabschätzung der Riemann–Siegel-Formel*, Dissertation, Göttingen, 1979 (reissued 2015); full asymptotic expansion of $R(\tau)$ in $\tau^{-1/2}$ with 1-periodic real-analytic coefficients.
- [3] M.V. Berry, *The Riemann–Siegel expansion for the zeta function: high orders and remainders*, Proc. R. Soc. Lond. A **450** (1995), 439–462.
- [4] M.V. Berry and J.P. Keating, *A new asymptotic representation for $\zeta(\frac{1}{2} + it)$ and quantum spectral determinants*, Proc. R. Soc. Lond. A **437** (1992), 151–173.
- [5] H.M. Edwards, *Riemann’s Zeta Function*, Academic Press, 1974, §7.4 (Riemann–Siegel contour-integral representation).
- [6] L. Hörmander, *The Analysis of Linear Partial Differential Operators I*, Springer, 1983, Theorem 7.3.1 (Schwartz–Paley–Wiener).
- [7] E.M. Stein, *Harmonic Analysis: Real-Variable Methods, Orthogonality, and Oscillatory Integrals*, Princeton, 1993, Chap. VIII §1 (non-stationary phase).